



CFC

CATS FOR CODES VCT DATABASE

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MONGODB

Reason we choose:

This is because mongodb flexibility offers a better solution then ordinary rigid relational schema where its not good for what we are require to build with stat teams roles and match history

Key Features

Flexibility

Nested data

Scalability

Next is nested data support allow player info match history and team history in a single document

there also scalability which allow future vot champions to be added and the Json like structure allowing easier management



Why it matters

Eliminates the need for intermediaries and builds trust.





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DATABASE SYSTEM DESIGN AND ARCHITECTURE.



Architecture used

Client → Application →
Database

Client (User/Terminal)

User interacts with terminal, in which
this is where commands are
executed to insert and retrieve data.

Application Layer (Redis CLI / Backend)

This layer processes user
commands and communicates
with the database.

Database Layer

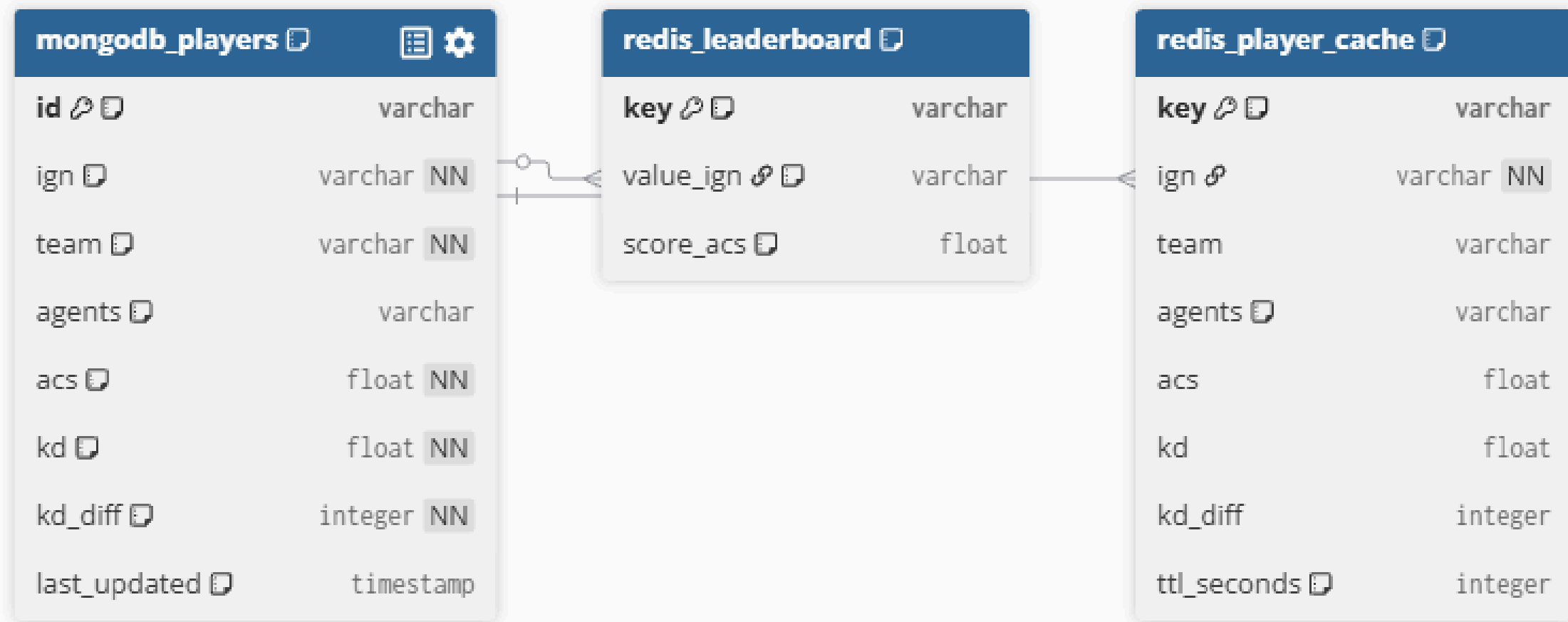
The database layer uses Redis to
store data in memory. The reason on
why Redis was chosen because it
provides fast read/write operations
and supports flexible NoSQL
structures.





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ERD DIAGRAM



#	key	value_ign	score_acs
1	vct:leaderboard:acs	TenZ	245.8
2	vct:leaderboard:acs	aspas	262.1
3	vct:leaderboard:acs	something	255.3
4	vct:leaderboard:acs	f0rsakeN	231.7
5	vct:leaderboard:acs	ZmjjKK	268.4

public.mongodb_players [Edit](#) [Download](#)

#	id	ign	team	agents	acs	kd	kd_diff	last_updated
	varchar	varchar not_null unique	varchar not_null	varchar	float not_null	float not_null	integer not_null	timestamp
1	65f12a3b4c5d6e7f8a9b0c11	TenZ	Sentinels	Jett, Omen, Reyna	245.8	1.22	142	now()
2	65f12a3b4c5d6e7f8a9b0c12	aspas	Leviatán	Jett, Raze, Neon	262.1	1.38	289	now()
3	65f12a3b4c5d6e7f8a9b0c13	something	Paper Rex	Jett, Reyna, Breach	255.3	1.28	210	now()
4	65f12a3b4c5d6e7f8a9b0c14	f0rsakeN	Paper Rex	Yoru, Breach, Cypher	231.7	1.14	88	now()
5	65f12a3b4c5d6e7f8a9b0c15	ZmjjKK	EDward Gaming	Jett, Neon, Chamber	268.4	1.31	254	now()

#	key	ign	team	agents	acs	kd	kd_diff	ttl_seconds
	varchar	varchar not_null	varchar	varchar	float	float	integer	integer
1	player:cache:tenz	TenZ	Sentinels	["Jett", "Omen", "Reyna"]	245.8	1.22	142	60
2	player:cache:aspas	aspas	Leviatán	["Jett", "Raze", "Neon"]	262.1	1.38	289	60
3	player:cache:forsaken	f0rsakeN	Paper Rex	["Yoru", "Breach", "Cypher"]	231.7	1.14	88	60
4	player:cache:zmjjkk	ZmjjKK	EDward Gaming	["Jett", "Neon", "Chamber"]	268.4	1.31	254	60





EXAMPLE

evan showing aspas his stats



```
127.0.0.1:6379> HSET player:1 name "TenZ" team "Sentinels" role "Duelist" agents "Jett,Raze" ACS 265 KD 1.32 HS 24 KILLS 240
127.0.0.1:6379> HSET player:2 name "Derke" team "Fnatic" role "Duelist" agents "Jett,Raze" ACS 270 KD 1.30 HS 26 KILLS 235
127.0.0.1:6379> HSET player:3 name "førsakeN" team "PRX" role "Flex" agents "Yoru,Neon,Jett" ACS 255 KD 1.25 HS 23 KILLS 220
127.0.0.1:6379> HSET player:4 name "aspas" team "LOUD" role "Duelist" agents "Jett,Raze" ACS 275 KD 1.35 HS 25 KILLS 245
127.0.0.1:6379> HSET player:5 name "yay" team "NRG" role "Chamber" agents "Chamber,Jett" ACS 260 KD 1.28 HS 27 KILLS 230
(integer) 8
127.0.0.1:6379> HSET player:6 name "something" team "PRX" role "Duelist" agents "Jett" ACS 268 KD 1.31 HS 22 KILLS 238
127.0.0.1:6379> HSET player:7 name "Leo" team "Fnatic" role "Initiator" agents "Sova,Fade" ACS 240 KD 1.20 HS 28 KILLS 210
127.0.0.1:6379> HSET player:8 name "Chronicle" team "Fnatic" role "Flex" agents "Viper,Brimstone" ACS 238 KD 1.18 HS 24 KILLS 205
127.0.0.1:6379> HSET player:9 name "nAts" team "Liquid" role "Sentinel" agents "Cypher,Viper" ACS 235 KD 1.22 HS 29 KILLS 200
(integer) 8
127.0.0.1:6379> HSET player:10 name "ZmjjKK" team "EDG" role "Duelist" agents "Jett" ACS 272 KD 1.33 HS 23 KILLS 242
(integer) 8
127.0.0.1:6379> HGETALL player:4
```

```
1) "name"
2) "aspas"
3) "team"
4) "LOUD"
5) "role"
6) "Duelist"
7) "agents"
8) "Jett,Raze"
9) "ACS"
10) "275"
11) "KD"
12) "1.35"
13) "HS"
14) "25"
15) "KILLS"
16) "245"
```

```
127.0.0.1:6379> HGETALL player:6
1) "name"
2) "something"
3) "team"
4) "PRX"
5) "role"
6) "Duelist"
7) "agents"
8) "Jett"
9) "ACS"
10) "268"
11) "KD"
12) "1.31"
13) "HS"
14) "22"
15) "KILLS"
16) "238"
```

```
127.0.0.1:6379> HGETALL player:7
1) "name"
2) "Leo"
3) "team"
4) "Fnatic"
5) "role"
6) "Initiator"
7) "agents"
8) "Sova,Fade"
9) "ACS"
10) "240"
11) "KD"
12) "1.20"
13) "HS"
14) "28"
15) "KILLS"
16) "210"
```

aspas result code

something



Timeline

Week 1 (27-2) - Project
concept and build

Week 2 (3-9) -Building
and running

week 3 (10-15) - Quailty
control and finalization

Task distribution

Evan - Data base schema
design

Ziyad - Data input and
source

Lando - Quaries and
back end

Risk management

1. Lack of time =
efficient time usage
and increase
productivity

2. Data inconsistentcy
= use predefined data
schema format

3. Technical issue =
switching to other db

PROJECT MANAGEMENT



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THANK YOU



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